CSCE-612 HW 2 Report

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1. random0.irl ++ invalid record: jump into fixed DNS header

```
C:\Users\Justus\Downloads>.\HW2.exe random0.irl 128.194.135.82
Lookup   : random0.irl
Query    : random0.irl, type 1, TXID 0x4425
Server   : 128.194.135.82
********************************
Attempt 0 with 29 bytes... response in 38 ms with 82 bytes
  TXID 0x4425 flags 0x8400 questions 1 answers 2 authority 0 additional 0
  succeeded with Rcode = 0
  ------------ [questions] ----------
  random0.irl type 1 class 1
  ------------ [answers] ------------
  ++ invalid record: jump into fixed DNS header
```

random3.irl Timeout

```
C:\Users\Justus\Downloads>.\HW2.exe random3.irl 128.194.135.82
Lookup   : random3.irl
Query    : random3.irl, type 1, TXID 0xBCAD
Server   : 128.194.135.82
********************************
Attempt 0 with 29 bytes... timeout in 10007 ms
Attempt 1 with 29 bytes... timeout in 10017 ms
Attempt 2 with 29 bytes... timeout in 10017 ms
```

random5.irl ++ invalid record: jump beyond packet boundary

```
C:\Users\Justus\Downloads>.\HW2.exe random5.irl 128.194.135.82
Lookup   : random5.irl
Query    : random5.irl, type 1, TXID 0x6B82
Server   : 128.194.135.82
********************************
Attempt 0 with 29 bytes... response in 38 ms with 71 bytes
  TXID 0x6B82 flags 0x8400 questions 1 answers 2 authority 0 additional 0
  succeeded with Rcode = 0
  ------------ [questions] ----------
  random5.irl type 1 class 1
  ------------ [answers] ------------
  random.irl A 1.1.1.1 TTL = 30
  ++ invalid record: jump beyond packet boundary
```

random6.irl ++ invalid record: jump loop

```
C:\Users\Justus\Downloads>.\HW2.exe random6.irl 128.194.135.82
Lookup   : random6.irl
Query    : random6.irl, type 1, TXID 0xAE71
Server   : 128.194.135.82
********************************
Attempt 0 with 29 bytes... response in 32 ms with 59 bytes
  TXID 0xAE71 flags 0x8400 questions 1 answers 2 authority 0 additional 0
  succeeded with Rcode = 0
  ------------ [questions] ----------
  random6.irl type 1 class 1
  ------------ [answers] ------------
  ++ invalid record: jump loop
  random6.irl CNAME ± TTL = 30
  ++ invalid record: jump loop
```

2. random1.irl ++ invalid section: not enough records

```
C:\Users\Justus\Downloads>.\HW2.exe random1.irl 128.194.135.82
Lookup   : random1.irl
Query    : random1.irl, type 1, TXID 0x775A
Server   : 128.194.135.82
********************************
Attempt 0 with 29 bytes... response in 6 ms with 468 bytes
  TXID 0x775A flags 0x8600 questions 1 answers 1 authority 0 additional 65535
  succeeded with Rcode = 0
  ------------ [questions] ----------
  random1.irl type 1 class 1
  ------------ [answers] ------------
  random.irl A 1.1.1.1 TTL = 30
  ------------ [authority] ----------
  ------------ [additional] ---------
  Episode.IV A 2.2.2.2 TTL = 30
  A.NEW.HOPE A 2.2.2.2 TTL = 30
  It.is.a.period.of.civil.war A 2.2.2.2 TTL = 30
  Rebel.spaceships A 2.2.2.2 TTL = 30
  striking.from.a.hidden.base A 2.2.2.2 TTL = 30
  have.won.their.first.victory A 2.2.2.2 TTL = 30
  against.the.evil.Galactic.Empire A 2.2.2.2 TTL = 30
  During.the.battle A 2.2.2.2 TTL = 30
  Rebel.spies.managed A 2.2.2.2 TTL = 30
  to.steal.secret.plans A 2.2.2.2 TTL = 30
  to.the.Empires.ultimate.weapon A 2.2.2.2 TTL = 30
  ++ invalid section: not enough records
```

3. random7.irl ++ invalid section: not enough records

```
C:\Users\Justus\Downloads>.\HW2.exe random7.irl 128.194.135.82
Lookup   : random7.irl
Query    : random7.irl, type 1, TXID 0x74ED
Server   : 128.194.135.82
********************************
Attempt 0 with 29 bytes... response in 4 ms with 42 bytes
  TXID 0x74ED flags 0x8400 questions 1 answers 2 authority 0 additional 0
  succeeded with Rcode = 0
  ------------ [questions] ----------
  random7.irl type 1 class 1
  ------------ [answers] ------------
  ++ invalid record: truncated jump offset
  random7.irl CNAME M TTL = 30
  ++ invalid section: not enough records
```

4. random4.irl ++ invalid record: truncated name

```
C:\Users\Justus\Downloads>.\HW2.exe random4.irl 128.194.135.82
Lookup   : random4.irl
Query    : random4.irl, type 1, TXID 0xFD65
Server   : 128.194.135.82
********************************
Attempt 0 with 29 bytes... response in 5 ms with 34 bytes
  TXID 0xFD65 flags 0x8400 questions 1 answers 1 authority 0 additional 11
  succeeded with Rcode = 0
  ------------ [questions] ----------
  random4.irl type 1 class 1
  ------------ [answers] ------------
  ++ invalid record: truncated name
```

random4.irl ++ invalid record: RR value length stretches the answer beyond packet

```
C:\Users\Justus\Downloads>.\HW2.exe random4.irl 128.194.135.82
Lookup   : random4.irl
Query    : random4.irl, type 1, TXID 0xA730
Server   : 128.194.135.82
********************************
Attempt 0 with 29 bytes... response in 6 ms with 421 bytes
  TXID 0xA730 flags 0x8400 questions 1 answers 1 authority 0 additional 11
  succeeded with Rcode = 0
  ------------ [questions] ----------
  random4.irl type 1 class 1
  ------------ [answers] ------------
  random.irl A 1.1.1.1 TTL = 30
  ------------ [authority] ----------
  ------------ [additional] ---------
  Episode.IV A 2.2.2.2 TTL = 30
  A.NEW.HOPE A 2.2.2.2 TTL = 30
  It.is.a.period.of.civil.war A 2.2.2.2 TTL = 30
  Rebel.spaceships A 2.2.2.2 TTL = 30
  striking.from.a.hidden.base A 2.2.2.2 TTL = 30
  have.won.their.first.victory A 2.2.2.2 TTL = 30
  against.the.evil.Galactic.Empire A 2.2.2.2 TTL = 30
  During.the.battle A 2.2.2.2 TTL = 30
  Rebel.spies.managed A 2.2.2.2 TTL = 30
  ++ invalid record: RR value length stretches the answer beyond packet
```

random4.irl ++ invalid record: truncated RR answer header

```
C:\Users\Justus\Downloads>.\HW2.exe random4.irl 128.194.135.82
Lookup   : random4.irl
Query    : random4.irl, type 1, TXID 0xFCAA
Server   : 128.194.135.82
********************************
Attempt 0 with 29 bytes... response in 5 ms with 212 bytes
  TXID 0xFCAA flags 0x8400 questions 1 answers 1 authority 0 additional 11
  succeeded with Rcode = 0
  ------------ [questions] ----------
  random4.irl type 1 class 1
  ------------ [answers] ------------
  random.irl A 1.1.1.1 TTL = 30
  ------------ [authority] ----------
  ------------ [additional] ---------
  Episode.IV A 2.2.2.2 TTL = 30
  A.NEW.HOPE A 2.2.2.2 TTL = 30
  It.is.a.period.of.civil.war A 2.2.2.2 TTL = 30
  Rebel.spaceships A 2.2.2.2 TTL = 30
  ++ invalid record: truncated RR answer header
```

5. I think random8.irl starts by correctly generating a response, until it gets to the additional records section. The number of additional records is always set to 11. Then for the actual additional records, a random number [0, 11] of phrases from the Star Wars opening scene are made into records, and returned in the additional records section. Then, one of the following errors is applied to this:
    a. Only the first N bytes of the response are sent, yielding ++ invalid record: truncated name.
    b. A record with a bunch of “lol”s is inserted, overflowing the buffer, resulting in ++ invalid record: truncated RR answer header.
    c. The length of a random record is set to a larger number, resulting in a ++ invalid record: RR value length stretches the answer beyond the packet.